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A Critical Review of Direct Detection in the Search for Dark Matter

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Abstract

We present a comparative assessment of the leading direct detection experiments for Dark Matter (DM), based on a unified framework for detector sensitivity. Using a toy model, we identify the four main drivers of detector sensitivity as exposure, target-material response, recoil-energy detection threshold, and background mitigation, and derive how each contributes to the observable cross-section limits. We assess and compare different detection techniques such as noble-liquid time projection chambers, cryogenic calorimeters, scintillation crystals, and superheated bubble chambers, based on their design choices and resultant performances in these drivers. Mapping these optimisation patterns onto the DM mass-cross-section parameter space, we identify the parameter regions where each technology is the most effective. We identify noble-liquid time projection chambers as the most promising path to improve on the global best spin-independent cross-section limit of 2.2×10^{-48} cm^2 in the ~ 40 GeV/c^2 mass region, and cryogenic calorimeters as the most promising in the < 10 GeV/c^2 mass range. We propose a review of the current direct detection experiment ensemble's ability to constrain the properties of remaining DM particle candidates in the GeV/c^2 mass range and discuss the future of the field of direct detection.

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Research and Writing Methods

The project began with a group discussion to consolidate our prior understanding of dark matter and to outline potential research directions. Following our initial meeting with our supervisor, the scope of the project was refined to focus specifically on direct detection experiments. We conducted our literature review using InspireHEP and arXiv to access review articles and primary publications from leading experimental collaborations, and we used the citation networks within these papers to identify additional relevant sources. As a team, we had weekly meetings to evaluate papers and identify emerging themes. These discussions, along with guidance from our supervisor, shaped the analytical focus of the project. All written material was produced by members of the group. Generative AI tools were used for minor grammatical, latex formatting improvements and helping to explain topics, and were not used to generate content, text, or any models or analysis presented. A github repository containing the code for models we developed can be found at [1], and was used to create all plots contained, unless expressed otherwise.

1 Introduction

This report aims to present a comparative, physics-driven review of the leading direct-detection experiments for dark matter (DM). The objective is to analyse how detector design choices determine each experiment's cross-section exclusion limits on the DM parameter space. To support this analysis, we develop a toy model that isolates key drivers of improved experimental sensitivity, providing a starting point to compare detectors of different design. Finally, we aim to map the sensitivity drivers onto the DM mass-cross-section parameter space, to determine where current experiments remain limited and to identify which drivers future experiments must prioritise to improve global exclusion limits.

Despite forming approximately 85% of matter content of the universe [2], the nature of DM is a profound open question in physics. A range of cosmological and astrophysical evidence, outlined in Section 2, suggests the existence of dark, non-baryonic matter. This non-luminous matter cannot be accounted for by the Standard Model of particle physics. Understanding and detecting this unseen component of the universe has become a central goal of both astrophysicists and particle physicists.

Hypotheses invoking new fundamental particles are not the only ones that have been proposed for the DM problem. Alternative non-particle explanations such as modified gravity theories like MOND [3] or astrophysical compact objects like primordial black holes [4] are able to explain some observations, but particle candidates for DM are far more successful in reproducing the full range of astrophysical and cosmological observations [5]. Consequently, much of the field's efforts are directed towards the search for one or more fundamental dark particles through three complementary methods: direct detection, indirect detection and collider searches. Direct detection stands out, owing to its relative model independence, maturity of experimental methods, and ability to probe a broad region of the DM mass range [6]. As such we will not consider indirect detection or collider searches further. To lay the groundwork for the report, we begin with a summary of astrophysical and cosmological evidence for particle candidates of DM.

2 Astrophysical and Phenomenological Motivation

The astrophysical and cosmological evidence for DM does more than point to the existence of non-luminous mass, by also constraining the type of particle which DM could be. Galaxy rotation curves, gravitational lensing and the Cosmic Microwave Background (CMB) temperature anisotropy collectively restrict the properties of a DM candidate, helping to minimise the DM parameter space.

Galaxy rotation curves were one of the earliest indications of DM [7]. Measurements of the orbital velocities of stars and gas in spiral galaxies show rotation speeds which are approximately constant at large radii. These observations oppose the decline in velocity predicted by the application of Newtonian dynamics to the visible mass distribution [8]. The baryonic component of a galaxy decreases with radius, therefore the enclosed visible mass of the galaxy is insufficient to cause the orbital velocities observed. High resolution measurements of rotation curves have demonstrated the need for an extended non-luminous mass halo surrounding each galaxy to provide the necessary gravitational potential to maintain these increased orbital velocities [7]. Fig. 1 shows a plot of NGC 6503 orbital velocities' plotted against distance from the galactic centre. Alongside a plot of the observed data, the rotation curves of individual components of

the galaxy, namely visible matter, gas and a dark-halo are computed and plotted. The rotation curve for visible matter alone, is clearly in disagreement with the observed rotation curve.

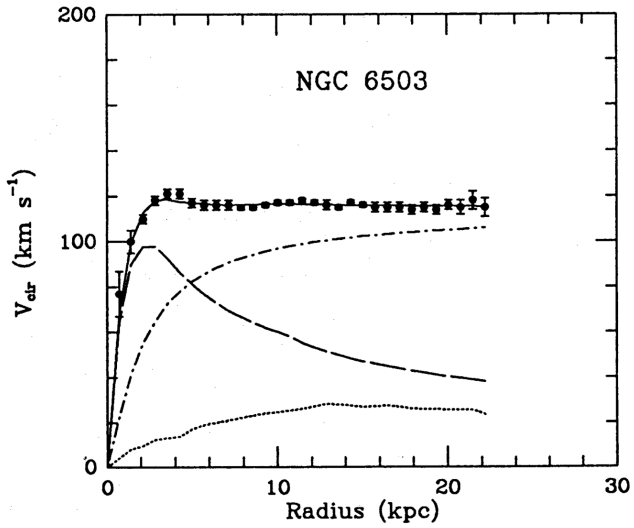


Figure 1: A rotation curve of NGC 6503 taken from [9]. The solid curve is a 3 parameter fit to the observed rotation curve. The dashed line shows the computed rotation curve of only visible matter. The dotted curve is computed for gas and the dash-dot curve computed for the dark halo.

Weak and Strong gravitational lensing of galaxies and galaxy clusters show that the lensing mass significantly exceeds the visible mass of the systems [10]. Reconstructing the mass distributions from lensing data shows large halos of matter not explained by baryonic matter alone. A reconstruction of the gravitational lensing mass of the merging cluster $1E0657-558$ [10] is shown in Fig. 2. X-Ray emissions shown in colour show the clustering of hot gas containing most baryonic matter, while the green contours show the weak lensing reconstruction. The gravitational-lensing mass peaks are displaced from the gas providing evidence for the existence of dark matter.

The temperature anisotropies of the CMB further demonstrate the need for non-baryonic matter. The angular power spectrum is sensitive to the densities of baryons, DM and radiation. Models that don't include DM fail to reproduce peaks in the spectrum. A cosmological model including cold dark matter (ΛCDM) fits well to measurements by the Planck satellite [11]. In Fig. 3, taken from [12] the Temperature power spectrum of CMB is plotted for varying dark matter density contributions between 0.11 to 0.43, holding all other parameters constant, shown with blue lines. The black line shows the best fit to the Planck data released in 2018 [11].

Together, these distinct pieces of evidence suggest that most of the matter in the universe is weakly interacting, non-baryonic and non-luminous motivating the search for DM.

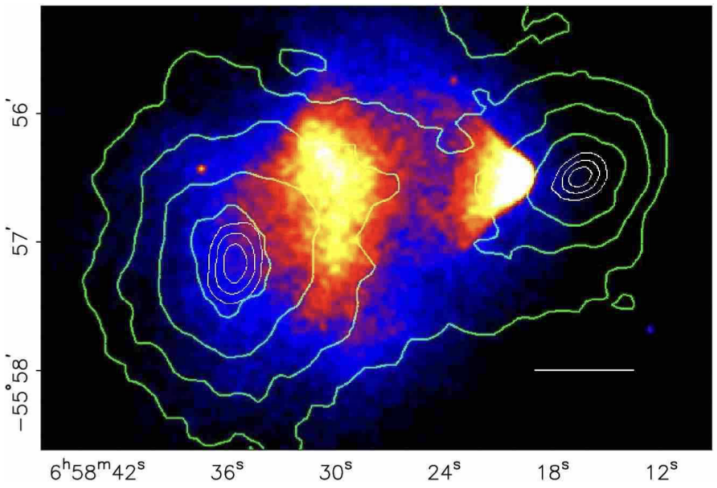


Figure 2: Image of the merging cluster taken from [10]. The green contours show the weak lensing reconstruction of the gravitational potential. The colour is the X-Ray emissions of clustering hot gas

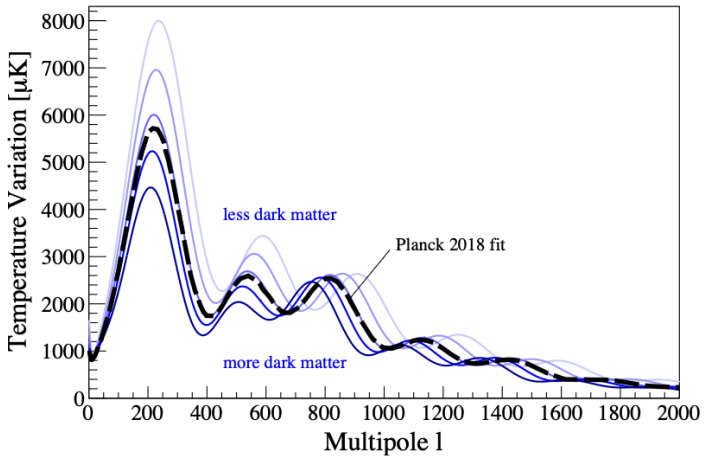


Figure 3: Temperature power spectrum of CMB from [12]. Blue lines show varying dark matter density contributions between 0.11 and 0.43. The black line shows the best fit to Planck data from 2018 [11]

3 Sensitivity Drivers

Direct detection experiments employ a wide range of different technologies, so comparing based off design and engineering obscures the underlying physics that drives the expected number of detectable DM events. Instead, we consider their ability to constrain the DM mass-cross-section parameter space. The standard framework for presenting these constraints is an exclusion plot, which shows DM mass m_χ on the x-axis and the DM-nucleon scattering cross section σ_N on the y-axis. On this plot the maximum σ_N consistent with the data measured at a given confidence level is indicated for each m_χ by a point on a curve that separates allowed regions from excluded regions. An experiment is said to be more sensitive for a given m_χ if it has a lower σ_N exclusion limit, meaning it can exclude weaker interaction strengths.

The aim of the following section is to implement a toy model that demonstrates the theoretical principles of direct detection, in order to physically motivate the key drivers of experimental sensitivity. These drivers will inform the analysis of specific detection techniques in Section 4.

3.1 Event Rates from First Principles

Direct detection analyses begin by assuming that the Earth moves through the galactic DM halo consisting of non-relativistic particles with relative speeds $\sim 10^{-3}c$ [13]. The rare scattering of DM particles with atomic nuclei creates a measurable recoil energy, and the expected number N_χ of these events provides the basis for constructing exclusion limits. N_χ can be expressed as:

$$N_\chi = M_T T_0 R \alpha, \tag{1}$$

where M_T is the fiducial mass of the detector, T is the live-time, R is the event rate, and α is a constant acceptance factor that represents the probability of a DM scattering event in the fiducial mass producing a signal. This term absorbs geometric fiducialisation (restriction to an inner fiducial volume) and an energy-independent efficiency, but does not include energy-dependent efficiency effects which are introduced later in Section 3.3.

To construct exclusion limits from N_χ, we determine the greatest N_χ that is consistent with the observed data. Since most modern experiments report limits using a Feldman-Cousins approach to confidence intervals [14], we adopt this approach throughout. We start by introducing a toy model with the following assumptions:

- The scattering rate is energy independent
- DM particles are incident with identical speeds v_0
- Zero signal rate from background processes
- DM-Nucleon interactions are only spin independent (SI)
- Scattering is always coherent

Under these assumptions, R is modelled as:

$$R = \frac{\rho_0 v_0}{m_\chi} \frac{N_A}{A_T} \left(\frac{\mu_T}{\mu_N}\right)^2 A_T^2 \sigma_N, \tag{2}$$

where ρ_0 is the local DM density, v_0 the characteristic speed, and $\mu_{T/N}$ the reduced mass of the DM-atom and DM-nucleon systems respectively. This results from combining the incident DM flux, the density of target nuclei, and the SI scattering probability, which scales as $\sim A_T^2$ for coherent scattering assuming the DM particle has approximately equal coupling to protons and neutrons [15], A_T is the nucleon number of the atom. Coherence holds when the DM de Broglie wavelength is greater than the target nuclear radius, so the nucleons contribute in phase. Thus combining Equations 1 and 2, with $N_{90\%} = 2.3$, we obtain:

$$\sigma_{N^{90\%}} = \frac{N_{90\%} m_\chi \mu_n^2}{\rho_0 v_0 M_T T_0 N_A A_T \mu_A^2 \alpha}. \tag{3}$$

We plot this relationship for our toy model below in Fig. 4:

From Eq. 3 we see that the exclusion limit scales linearly with exposure ($M_T T_0$) and A_T, identifying the dominant sensitivity drivers of this simplified model. Figure 4 shows that the toy model reproduces the qualitative shape of the exclusion limit, it differs by up to an order of magnitude from the LZ 2025 results [16]. The toy model exhibits incorrect curvature in the low-mass region arising from its simplified assumptions including, energy-independent scattering, ignoring detector effects and a zero event background. Incorporating these effects is necessary to improve the expected sensitivity and is explored in the following subsections

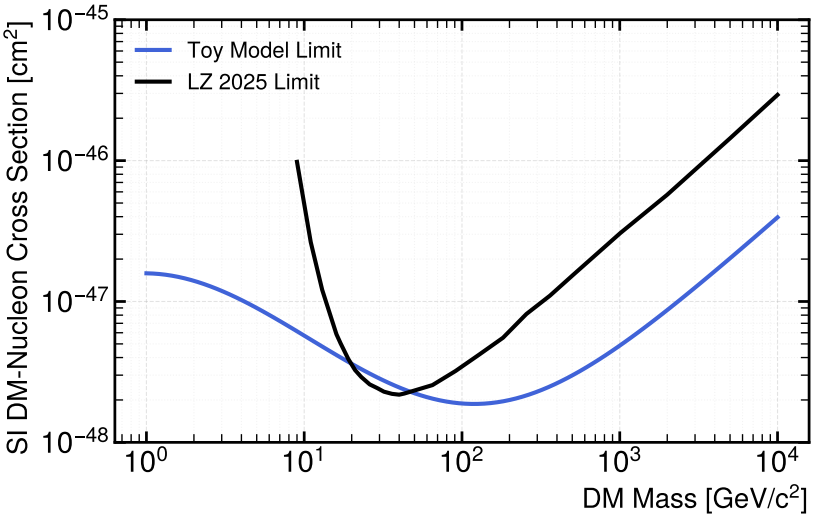


Figure 4: Plot of Eq. 3 showing $\sigma_{n^{90\%}}$ as a function of m_χ, computed using the code available in our github repository [1]. 2025 LUX-ZEPLIN (LZ) results [16] are overlayed in black. Astrophysical and detector constants taken as follows: $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$, $v_0 = 220\,\mathrm{km\,s^{-1}}$, $M_T = 4200\,\mathrm{kg}$, $A_T = 131\,\mathrm{g\,mol^{-1}}$, $T_0 = 1\,\mathrm{year}$ and $\alpha = 0.5$.

3.2 Energy-dependent Rate and Nuclear Response

The toy model assumes an energy-independent scattering rate, but real detectors measure from a distribution of nuclear recoil energies [13]. To model the distribution of nuclear recoil energies, we consider the scattering of a DM particle, χ, off a target nucleus, N,

$$\chi + N \to \chi + N. \tag{4}$$

In the lab frame, the target nuclei are stationary, applying conservation of energy and momentum in the non-relativistic limit, the recoil energy is:

$$E_R = \frac{\mu_T^2 v^2 (1 - \cos\theta)}{m_T}, \tag{5}$$

v is the incident velocity of the DM particle, θ the scattering angle and m_T the mass of the target nuclei. For a given recoil energy, only DM particles above a minimum velocity:

$$v_{min} = \sqrt{\frac{m_T E_R}{2\mu_T^2}}, \tag{6}$$

are able to generate a detectable signal.

The differential rate follows from integrating the differential scattering probability over the velocity distribution [13]:

$$\frac{dR}{dE_R} = N_T \frac{\rho_0}{m_\chi} \int_{v_{min}}^{v_{max}} d^3v f(\boldsymbol{v}) v \frac{d\sigma}{dE_R}, \tag{7}$$

where N_T is the number of target particles and $f(\boldsymbol{v})$ is the galactic DM velocity distribution boosted into the Earth's frame by velocity v_E.

For spin-independent (SI) interactions the scattering is isotropic and changing variables from θ to E_R gives:

$$\frac{d\sigma}{dE_R} = \frac{m_T \sigma_T}{2\mu_T^2 v^2} F^2(E_R), \tag{8}$$

where σ_T is the total DM-nucleus cross section and $F(E_R)$ the nuclear form factor.

In contrast to SI scattering, spin-dependent (SD) scattering couples to nuclear spin rather than nucleon number. For SD scattering, coherence across all nucleons doesn't occur and thus SD cross sections lack A_T^2 scaling, instead depending on spin expectation values [15, 17]. As a direct consequence, only target nuclei with non-zero nuclear spin contribute to SD scattering. This section will not discuss SD scattering further and as such will take $\sigma_{N^{90\%}}$ to be the SI scattering cross-section.

Returning to SI scattering, $F(E_R)$ accounts for the nucleus being non-point like and thus at large momentum transfers, scattering loses coherence [17, 18]. Physically, as the de Broglie wavelength,

$$\lambda = \frac{2\pi}{\sqrt{2m_T E_R}}, \tag{9}$$

of the interaction becomes $\lesssim r$, the nuclear radius ($r \sim A_T^{1/3}$), then scattering is no longer in phase across the nucleons, and the cross section is reduced. In this report we adopt a phenomenological model for nuclear form factor, the Helm form factor [18], $F_H(E_R)$. Combining Eq. 7 and Eq. 8 yields:

$$\frac{dR}{dE_R} = N_T \frac{\rho_0}{m_\chi} \frac{m_T \sigma_T}{2\mu_T^2} F_H^2(E_R) \int_{v_{min}}^{v_{max}} \frac{f(\mathbf{v})}{v} d^3v. \tag{10}$$

By modelling $f(\boldsymbol{v})$ as a Maxwell distribution truncated at the galactic escape velocity and boosted by v_E [13] we are able to numerically compute $\sigma_{90\%}$ for an event rate where the nucleus is considered point like ($F^2(E_R) = 1$) and an event rate with the Helm form factor using Xenon as a target. The results are compared against the toy model and shown in Fig. 5.

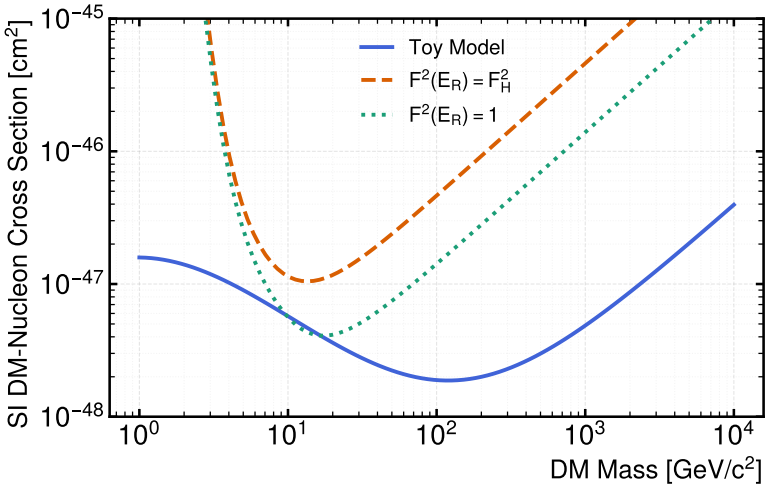


Figure 5: The figure shows a comparison of exclusion plots; the toy model, an energy dependent rate with $R(E_R) \propto F^2 = 1$ and an energy dependent rate, $R(E_R) \propto F_H^2(E_R)$. The differential rates were integrated over the E_R window, 5 keV $\leq E_R \leq$50 keV, all parameters appearing in both Eq. 3 and Eq. 7 take the same values as in Fig. 4

The largest difference between the exclusion limits in Fig. 5 is seen for low DM masses $m_\chi \lesssim 10\,\mathrm{GeV/c^2}$, where only particles that exceed $v \geq v_{min}(E_R)$ have sufficient energy to cause a measurable recoil. The effect becomes more pronounced as DM mass decreases, as a smaller

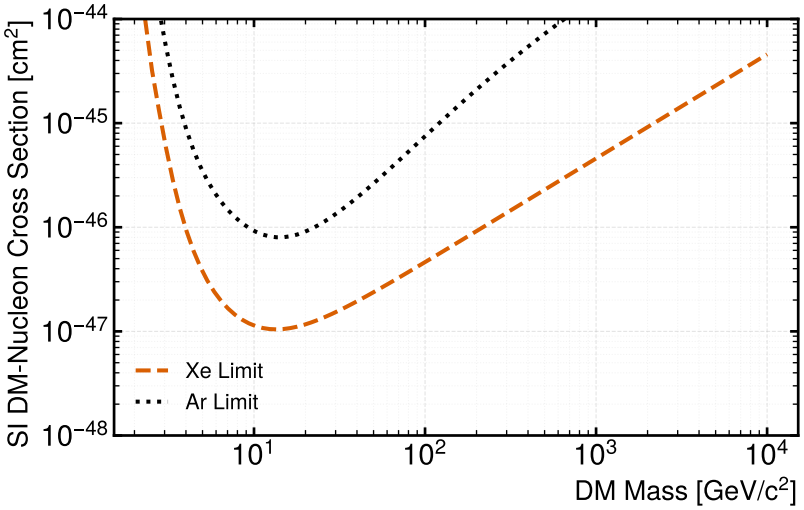


Figure 6: Comparison of $\sigma_{90\%}$ for two detectors with target materials of Xe and Ar respectively. Values of $\sigma_{90\%}$ computed using the energy dependent-recoil model, with energy dependent rate $R(E_R) \propto F_H^2(E_R)$. All parameters independent of A_T take the same values as in Fig. 4.

proportion of the Maxwell velocity distribution contributes to the count rate, causing the steep increase in σ_N. For high DM masses $m_\chi \gtrsim 30\,\mathrm{GeV}/c^2$, DM number density decreases as $\frac{\rho_0}{m_\chi}$, reducing the event rate and decreasing sensitivity. The toy model would be optimised for intermediate masses $10\,\mathrm{GeV} \lesssim m_\chi \lesssim 30\,\mathrm{GeV}/c^2$. Figure 5 also demonstrates the effect of the nuclear form factor on DM masses $\gtrsim 10\,\mathrm{GeV}/c^2$ where sensitivity has decreased because $\lambda \lesssim r$ as higher mass particles have larger kinetic energies and as a result coherence across the nucleus has been lost.

The coherence of scattering across the nucleus has a significant impact on detector sensitivity and is directly tied to the detector target's nuclear radius. The approximate relationship between r and A_T is $r \sim A_T^{1/3}$ [19], therefore we see that larger A_T will lead to greater coherence at higher recoil energies and as a result reduce the loss of sensitivity in higher mass regions. To demonstrate this effect in Fig. 6, we compare two theoretical detectors with differing target materials, Xe and Ar holding all other parameters constant as in the figures above.

3.3 Detector Energy Thresholds and Efficiency

We will now consider the detector energy thresholds which govern the window the differential rate is integrated over. The following section will explain the physics behind the measurement of recoil energy and demonstrate the effect the energy thresholds of a detector have on the scattering sensitivity, particularly in the low mass region.

The recoil energy is not directly measured in any type of DM direct detection experiment; instead, the recoil energy is inferred through measurements of quanta [6, 12], secondary excitations such as scintillation, ionisation or athermal phonons [20, 21]. The energy recoil is dissipated in two types of process: nuclear and electronic stopping [22]. Nuclear stopping transfers the recoil energy as kinetic motion. Electronic stopping leads to the secondary excitations outlined above. At low recoil energies the fraction of energy dissipated through electron stopping is insufficient to be distinguished from background by the detector [20], the corresponding recoil energy this occurs at is the lower energy threshold, $E_{R_{min}}$. The upper energy threshold is set because DM-Nucleon recoils are suppressed at high recoil energies by kinematics and nuclear

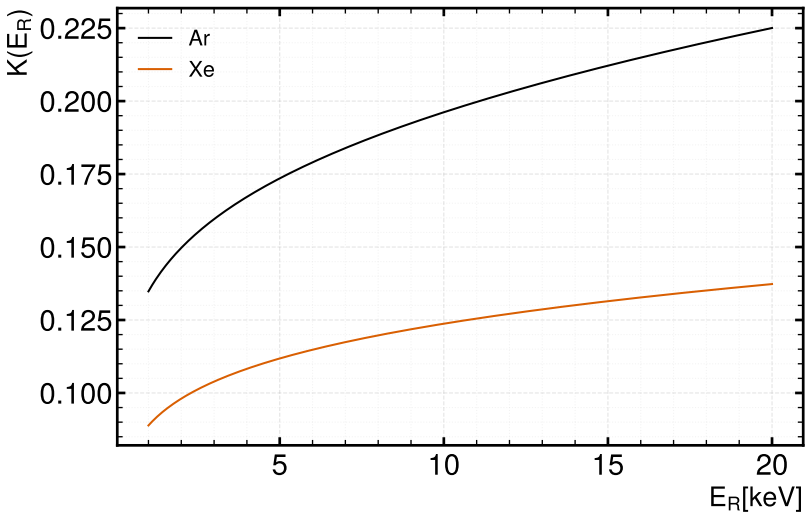


Figure 7: A plot comparing the relative fractions of electronic stopping, $K(E_R)$, for two detectors composed of Xe and Ar respectively. Parameters for the Lindhard model [22] were taken from [20]

form factor, while background events continue to rise. Therefore, including higher recoil energies reduces the signal-background ratio and weakens sensitivity. The fraction of total recoil energy dissipated by electronic stopping, $k(E_R)$, is given by:

$$K(E_R) = \frac{S_e(E_R)}{S_e(E_R) + S_n(E_R)}, \tag{11}$$

where S_e and S_n are the amounts of energy dissipated through electronic and nuclear stopping respectively. The amount dissipated in each channel is dependent on both E_R and the detector material. Since $E_{R,\min}$ is the largest driver of low mass sensitivity, it is important to understand how materials' fraction of electronic stopping changes as a function of E_R. This is well modelled by [22]:

$$K(E_R) = \frac{kg(\epsilon)}{1 + kg(\epsilon)}, \tag{12}$$

where, k is a material dependent parameter experimentally measured, $g(\epsilon)$ is an empirical fit to stopping power calculations and ϵ is an energy parameter modified by the nuclear number of the detector material. The exact forms of k, g and ϵ will not be expressed, but a full treatment is given in [20]. In Fig. 7 we compare $K(E_R)$ for two detectors composed of Xe and Ar respectively. The increased electronic stopping fraction of the Ar detector compared to the Xe detector would allow it to probe a lower DM mass region because it is more sensitive to smaller E_R due to its reduced $E_{R,\min}$. The parameter, $E_{R,\min}$, is the most significant driver of DM direct detection in mass regions $m_\chi \lesssim 10\ GeV/c^2$. We demonstrate the effect of reducing $E_{R,\min}$ in Fig. 8. As $E_{R,\min}$ is lowered the global sensitivity improves, and the maximum sensitivity is translated left to lower m_χ. This highlights the importance of small $E_{R,\min}$ in low mass searches, which can be achieved through detector material choices or improved statistical methodologies which allow better discrimination of $E_{obs} = K(E_R)E_R$ from background at low E_R.

3.4 Experimental Uncertainties

Direct detection sensitivities are subject to multiple sources of uncertainty arising from astrophysical parameters, detector response, and background processes. These uncertainties affect

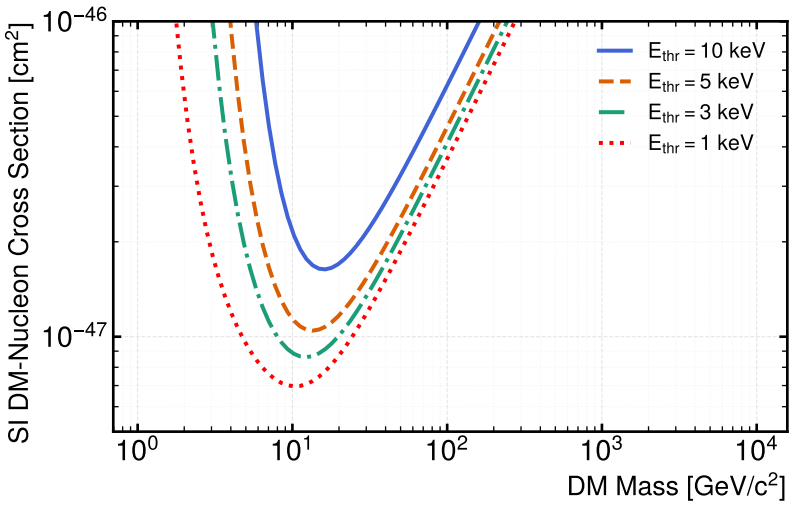


Figure 8: Plot showing the effect of reducing $E_{R,\min}$. As in other figures all other parameters take their previous values. The green line, corresponding to $E_{R,\min} = 5\,\text{keV}$, acts as a reference curve since it is present in Fig. 5 and Fig. 6.

both the normalisation and shape of exclusion plots [23]. In this section, we summarise significant sources of uncertainty applicable to all detector classes.

Astrophysical Uncertainty

As demonstrated in Section 3.2, the differential rate and thus the DM-nucleon cross section are dependent on ρ_0, $f(\boldsymbol{v})$, v_0 and v_{esc}. Variations in ρ_0 rescale exclusion plots as it is directly proportional to the scattering rate, while v_0 and v_{esc} affect the high-velocity tail of the distribution, which predominantly impacts the low-mass DM sensitivity. However, these uncertainties are dominated by the uncertainty in the form of $f(\boldsymbol{v})$ [24, 25]. Departures from the Maxwell distribution used in this report can significantly modify $R(E_R)$ especially in the region $m_\chi \lesssim 10\,\text{GeV}/c^2$ where only the high-velocity tail of $f(\boldsymbol{v})$ contributes due to $E_{R,\min}$ cutting off signals where $E_R \leq E_{R,\min}$. A full treatment of perturbations or alternative versions to the Maxwell distribution is beyond the scope of this report. Instead the remainder of this section will focus on uncertainties arising from detector response and background processes.

Detector Energy-Resolution Uncertainty

Real detectors are not perfect and don't have finite energy resolution. Instead the finite resolution broadens the recoil spectrum at all energies [13]. The impact of this smearing is most pronounced for small E_R. In this energy region, finite resolution causes $E_R \geq E_{R,\min}$ to be smeared below the energy threshold, reducing the observable rate at all mass ranges. The effect of smearing is most significant in the low mass region, where more events are closer to $E_{R,\min}$, and thus sensitivity is lost in this region. As outlined in Section 3.3, detectors measure quanta and as E_R decreases the number of quanta produced sharply falls. The reduction in the number of quanta introduces a counting statistic [20]. By treating the number of detected quanta as a Poisson-distributed variable with a mean, n_q, given by the Linhard Yield:

$$n_q = \frac{E_{obs}}{W}, \tag{13}$$

where W is the empirically-determined average energy required to produce a quantum in the material [20]. Convolving the true differential rate with the Poisson distribution gives the

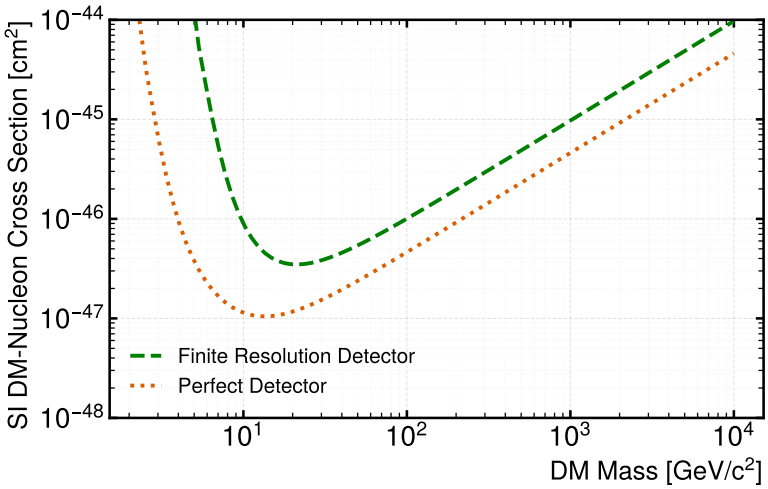


Figure 9: Plot compares a perfect detector to a detector with finite resolution modelled by a Poisson distribution [13]. The detector material is modelled as liquid Xe with $W = 13.7 \pm 0.2$eV [26]

observed differential rate and has a similar affect to raising $E_{R,\mathrm{min}}$. In Fig. 9 we compare the exclusion plots obtained from perfect energy resolution and smearing of signal quanta by a Poisson distribution. The effect of finite energy resolution is largest for $m_\chi \lesssim 10\,\mathrm{GeV}/c^2$ where a large fraction of events lie close to $E_{R,\mathrm{min}}$. Fluctuations in the detected quanta push events below the analysis window causing a reduction in sensitivity. Finite energy resolution therefore is a key experimental uncertainty in the low-mass region.

Background Processes

All direct detection experiments suffer from backgrounds which would obscure DM signals. These backgrounds fall into two broad categories: intrinsic backgrounds, originating from the detector and target material, and external backgrounds, coming from the surrounding environment. The categories are present in all direct detectors but their effects are experiment dependent.

Intrinsic backgrounds originate largely from materials that contain trace amounts of radioactivity, where the resulting decay products can produce signal-like events within the detector [27]. Therefore, radiopurity of detector materials is required to constrain this background process. These backgrounds typically define the baseline event rate.

External backgrounds consist primarily of cosmic rays and the following secondary interactions which can produce electronic and nuclear recoils [28]. Mitigating this background influences detector location, overburden and design of shielding in detectors. A further irreducible contribution to the external background comes from neutrinos. Due to the increase in sensitivity of modern direct detection experiments, coherent elastic neutrino-nucleus scattering ($CE\nu NS$) also provides a source of experimental uncertainty, often referred to as the "neutrino floor" [29]. Neutrinos incident on detectors become an unavoidable addition to the total recoil spectrum as they cannot be removed through detector design. The main issue of CEνNS is the overlap in recoil spectra with DM resulting in a degeneracy between signal and background [30]. Once the uncertainty in the neutrino background becomes comparable to the expected number of DM events the experiment shifts from signal to background-limited. This causes the scaling of sensitivity with exposure to become $\sigma_n^{90\%} \propto 1/\sqrt{M_T T_0}$ [31]. The transition to this regime is

dependent on energy thresholds and resolution as well as detector target materials and is not a hard limit on sensitivity. As a result, the "neutrino floor" is more accurately a "neutrino fog" [32] which introduces an irreducible statistical uncertainty. Improvements in neutrino-flux measurements, use of multi-target detectors, as CEνNS has a N^2 coherence factor ($N = A - Z$) instead of A^2 for DM [33], and an increase in spectral resolution can reduce the effects of the fog.

4 Experimental Implementations and Comparative Sensitivity

Having identified the key theoretical drivers of sensitivity in Section 3, our aim in this section is to assess how different experimental technologies optimise these drivers in practice. We focus on how the core design choices of each experiment type quantitatively shape its ability to maximise sensitivity across the $\mathcal{O}(0.1 - 10^3$ GeV/c$^2)$ range.

For clarity we recall the four drivers identified in Section 3, which form the basis for all comparisons in this section:

- Exposure ($M_T T_0$)
- Target-nucleus properties (A_T, Z, spin)
- Recoil-energy detection threshold / low-energy efficiency
- Background mitigation and suppression

We limit our analysis to four major experimental classes: Noble-liquid time projection chambers, cryogenic calorimeters, scintillation crystals, and superheated bubble chambers. While we may briefly mention other experiment types where relevant, the principles of these experiments map more clearly to the theoretical framework outlined in Section 3.1, so allow for a more consistent basis for our comparison.

4.1 Noble Liquid Time Projection Chambers

Noble-Liquid time projection chambers (TPCs) using liquid Xenon (LXe) or liquid Argon (LAr) are current frontrunners for SI WIMP searches for the mass range $m_\chi \gtrsim 10$ GeV/c^2. Dual phase detectors such as XENONnT, LUX-ZEPLIN (LZ), and PandaX-4T have a thin gaseous layer over the liquid target [34]. Single-phase TPCs, which operate without the gaseous layer, provide an alternative implemetation used in some LAr experiments. When WIMPs interact with target nuclei, recoil energy is deposited and is partitioned into three possible channels - scintillation, ionization and heat. Scintillation occurs when Xe_2* excimers formed by both excited and ground-state Xenon atoms de-excite and emit light of wavelength 178 nm [34]. Such scintillation light is detected as signal 1 (S1). Some deposited energy ionizes electrons in Xenon atoms, and the freed electrons are extracted and accelerated by electric fields in the detector into the gaseous layer at the top, where the second light signal (S2) is produced via electroluminescence [35]. In addition, the heat produced may be detected in the form of phonons by cryogenic bolometers [35]. The recoil energy can be reconstructed with the S1 and S2 signals, as well as photon and electron detection efficiencies [36]. The S1-S2 time-difference and S2 signal pattern [34] can be used to create a full 3D position reconstruction. This allows event selection to be restricted to the inner regions of the detector [35], which is shielded by the outer regions from radioactive backgrounds.

A defining strength of TPCs is their capacity to optimise exposure, by scaling to multi-tonne active masses. In the absence of shielding, external backgrounds would scale linearly with exposure. However, self-shielding effects in LXe lead to rapid attenuation of external radiation: LZ reports that approximately 2 cm of LXe reduces γ-ray backgrounds by an order of magnitude, while ~ 6 cm provides a similar suppression for radiogenic neutrons [37]. This allows a central fiducial region to be defined with reduced external background (geometric fiducialisation). After this geometric fiducialisation, external backgrounds fall exponentially with distance to the edge of the fiducial volume container [37]. This mitigates the usual degradation of statistical sensitivity from growing detector size, and allows increase in exposure without downside from background signals. meaning the usual limit on size from degraded statistical power is mitigated. Furthermore, the high densities of Ar (1.4 g/cm^3) and Xe (3 g/cm^3) allow large small volumes to contain even greater M_T of order $\mathcal{O}(6000$ kg) [38], and thereby maximise shielding effects and exposure simultaneously. Current leading instruments achieve fiducial masses of [39, 16, 40]:

$$M_T = 5.9\ t\ \text{(XENONnT)}, \qquad M_T = 5.5\ t\ \text{(LZ)}, \qquad M_T = 2.67\ t\ \text{(PandaX-4T)}$$

The advantageous background scaling does not continue indefinitely. Once external backgrounds are reduced by self-shielding, intrinsic backgrounds dominate the residual rate, and cannot be mitigated by additional shielding [37].

In LXe these intrinsic signals arise from ^{222}Rn daughter nuclei and trace ^{85}Kr; in LAr the limiting intrinsic background is ^{39}Ar $\beta-$decay [41]. These sources are uniformly distributed throughout the target volume [37], therefore scale linearly with M_T. Current fiducial masses represent the balance between maximising exposure and minimising intrinsic background contributions. As discrimination techniques improve, intrinsic background signals have lower statistical effect allowing setups with greater exposure. Next-generation detector XLZD is projected to reach σ_N at the neutrino floor with exposure of order $200 - 300$ tonne-years [42].

The choice of Xe or Ar as the target medium optimises different aspects of performance. Xenons high atomic mass $A_{\text{Xe}} = 131$, gives an initial advantage of $\sim A_{\text{Xe}}/A_{\text{Ar}} \approx 4$ over Argon due to the coherence of spin independent scattering as described in Section 3.2. Xe also consists of isotopes ^{129}Xe and ^{131}Xe with odd neutron number, allowing spin-dependant WIMP searches. Ar on the other hand is inexpensive and abundant compared to Xe, with Xe costing $\sim \$1M$ per tonne [43]. This feature gives LAr detectors a promising future, with the ability to scale to tens or hundreds of tonnes in fiducial mass as demonstrated by the upcoming Darkside-20k (20 t) and ARGO (300 t) [44]. In Fig. 10 we see exclusion limits for LXe and LAr experiments LUX and DarkSide-50 respectively.

LXe and LAr detectors also demonstrate differences in background mitigation strategies, as highlighted by the approximately two orders of magnitude difference in exclusion limits at $m_\chi \approx 10$ GeV/c^2 in Fig. 10. LZ is situated 1.5 km underground at Sanford Underground Research Facility [45], while DarkSide-50 operates at 1.4 km underground at Laboratori Nazionali del Gran Sasso [46]. Both experiments also place a large focus on dealing with trace radon gas, abundantly present in underground laboratories. LAr detectors make up for the lower A_T scaling through the improved discrimination capability. Ar scintillation shows strong separation between fast singlet and slow triplet emission. This enables pulse-shape discrimination with ER rejection $\sim 10^9$ [47], greater than that achievable only through S2/S1 discrimination in LXe.While LXe and LAr experiments are currently leading the field for the $m_\chi \gtrsim 10$ GeV/c^2 mass range, with lowest ever limits $\sigma_N = 2.2 \times 10^{-48}$cm^2 published by LZ on 01/07/2025 [16], this comes at the cost of high energy thresholds compared to cryogenic detectors. LXe TPCs

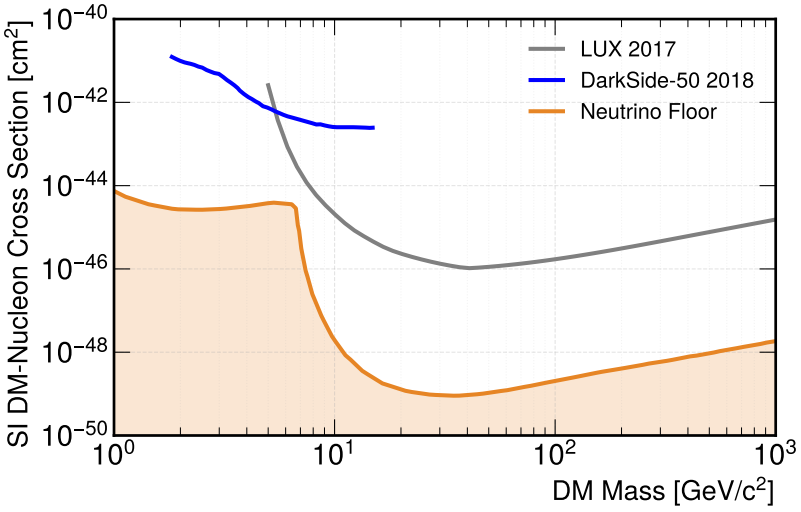


Figure 10: σ_N exclusion limits for LUX (2017) and DarkSide-50 (2018), demonstrated side-by-side with the neutrino floor. Run years chosen based on most recent year with comparable LXe and LAr results.

LZ and XENONnT operate with nuclear-recoil thresholds of $4-5$ keV in S1 + S2 analyses and for an S2 only search typically $\lesssim 1$ keV [39, 37].

The unique order in which the physical sensitivity drivers identified in Section 3 are optimised for dictates the parameter space in which experiments of this type operate optimally. While the world class exposures give the strongest limits around $m_\chi \sim$ 30 GeV/c^2, the high energy thresholds give a strong limit for their low mass sensitivity.

4.2 Cryogenic Calorimeters

Cryogenic detectors, such as those used by CRESST, EDELWEISS and SuperCDMS, are designed to identify WIMP recoils through measurement of phonons and ionisation in small, ultra-cold detectors. These experiments prioritise extremely low recoil-energy thresholds, capitalising on the mass range below the sensitivity range of noble liquid TPCs ($m_\chi \lesssim 10$ GeV/c^2).

The cryogenic detectors consist of target crystals (CaWO4 in the case of CRESST-II/III), as well as superconducting transition edge sensors (TESs) and light sensors. The detection modules operate at a temperature of 15 mK [48], just below the critical temperature of the TESs. A nuclear recoil deposits energy into the crystal, producing phonons which raise the temperature of the TESs. The transition from a superconductive to normal phase in the TESs allows a precise calorimetric measurement of the deposited energy. In addition to phonon detection, the corresponding light signal of the event due to scintillation in the CaWO4 crystal is detected by light sensors and simultaneously read out, which allows particle discrimination based on the light-yield ratio. This process is known as quenching. Electron-recoil events from γ/β backgrounds have a higher light-yield ratio than nuclear recoils [49].

The small size of detection modules reduces module heat capacities, which lowers CRESST's energy thresholds. The energy threshold of CRESST-II (whose cylindrical modules measure 40 mm for both diameter and height) is 0.3 keV, and allows the probe of dark matter particles with sub-GeV masses (down to 0.5 GeV/c^2 for spin-independent interactions). The sub-GeV range is dominated by scattering events from lighter nuclei (oxygen) as those from heavier nuclei (tungsten) have maximum deposited energies below the energy threshold [50]. For larger WIMP

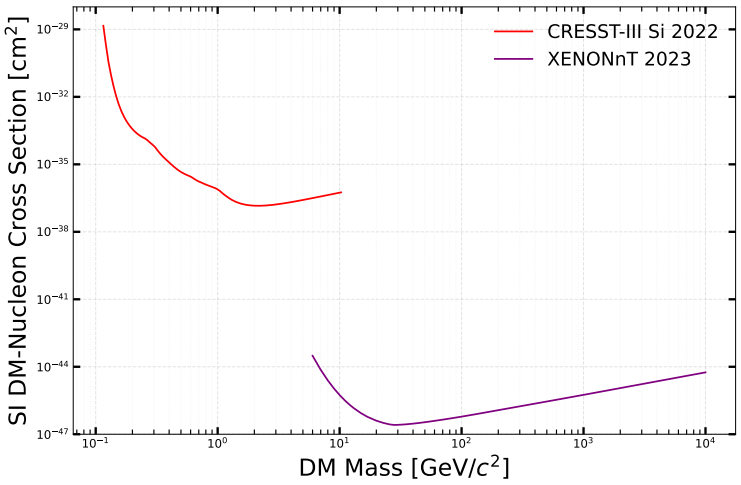


Figure 11: Plot showing σ_N exclusion limits for CRESST-III (2022) and XENONnT (2023).

masses, scattering events from tungsten become dominant as nucleus-WIMP cross sections scale with A_T^2. However, cross-section limits of CRESST-II for larger masses is higher than other LXe experiments (XENONnT) due to substantially smaller target mass and smaller exposure. A comparison of the exclusion limits of CRESST and XENONnT can be seen in Fig. 11.

The difference can in part be explained by cryogenic experiments relative difficulty to scale to larger exposures. Operating at $10 - 20$ mK imposes strict limits on the total thermal mass: at these temperatures the cooling power of dilution refrigerators is only $\mathcal{O}(\mu W)$ [48]. Increasing detector mass increases heat capacity of the system, thereby degrading threshold and resolution. TES also require low temperature response per unit energy, which prevents the use of larger crystals which have lower phonon-collection efficiency [51]. Consequently, cryogenic experiments rely on modularity, which causes heat load and complexity to rise faster than mass, presenting major engineering challenges. In practice cryogenic detectors are limited to exposures of $\mathcal{O}(1 - 10$ kg-years). As a result, XENONnT is able to operate with a 500 times greater exposure than CRESST-III, partially explaining the difference in sensitivity limits above $m_\chi = 10$ GeV/c^2 [52, 39].

Similar to LXe and LAr experiments, cryogenic calorimeters are placed in deep-underground laboratories (Gran Sasso for CRESST, SNOLAB for EDELWEISS), with active muon veto systems to reduce cosmogenic backgrounds. Another source of background signals are alpha decays on the crystal surface, which can mimic nuclear recoils. CRESST uses a fully scintillating housing to detect surface emission and improve the discrimination power in the analysis level [53].

The order of driver optimisation is clear in cryogenic calorimeters. Ultra-low energy thresholds ($\lesssim 1$ *keV*) are prioritised at the cost of an inability to scale to large exposures. While the low exposure is costly at higher mass ranges, effective background discrimination techniques allow experiments such as CRESST-III to make use of the low energy threshold and dominate in the $m_\chi \lesssim 10$ GeV/c^2 range.

4.3 Scintillation Crystals

Although scintillation-crystal detectors share the same aims as Noble liquid and cryogenic detectors, their can provide a strong complementary approach within direct detection. Experiments of this type, such as COSINE-100 and DAMA/LIBRA, consist of scintillation crystals as the target and photomultiplier tubes. When WIMPs deposit energy in the scintillation

crystals, electrons are excited to the conduction band and electron-hole pairs are created [54]. The energetic electrons are later captured by luminescent centres [54] (possibly a dopant, as in the thallium-doped NaI crystal used in COSINE-100), which would de-excite, thus emitting photons to be detected by photomultiplier tubes.

In principle modular NaI components of these experiments can be manufactured for low cost and in bulk. However, similar to cryogenic experiments their modular nature slightly limits coherent scaling of exposure, meaning their exposures are typically slightly lower than those of similar TPCs, with exposures ranging from 10 kg-yr to more than 1 tonne-year [55, 56, 57]. The usage of higher crystal mass is possible, but for scintillation crystals with a fixed radiopurity level, the increased mass raises background count rates. As a result, the experiment is only moderately optimised to maximise exposure. This also highlights the primary limitation of NaI-based searches. They are unable to discriminate effectively between Nuclear Recoil (NR) and Electron Recoil (ER) signals, and intrinsic contaminations of ^{40}K, ^{210}Pb, and ^{3}H dominate the low energy spectrum. Experiments mitigate these effects through radiopure crystal processing and liquid-scintillator veto systems.

One characteristic of these experiments is that non-proportionality between light emitted and energy deposited by an incident particle [58] can introduce further uncertainty. This places strong constraints on the energy threshold driver. Typical thresholds of $2-5$ keV can translate, after quenching, into NR thresholds of $10-20$ keV [59]; higher than both TPCs and cryogenic experiments. We would also expect the presence of both light (Na) and heavy (I) nuclei to allow strong sensitivity in both high and low m_χ regions of the DM parameter space. This is because Iodine nuclei benefit from the strong A_T scaling introduced in Section 3.1, and sodium recoils provide a lower energy threshold recoil channel. However, the quenching factors ($0.2-0.3$ for Na and $\lesssim 0.1$ for I), degrade the usable recoil energy spectrum due to the non-proportionality in light-yield described previously [59].

The combination of relatively good exposure, low thresholds and high target nucleon number positions these experiments uniquely to provide useful σ_N across the m_χ range of both TPC and cryogenic experiments. While their exclusion limits may be less restricting due to their fundamental limitations with respect to mitigating background signals, they are uniquely positioned for high-precision annual modulation searches. The stability of NaI scintillation over multiyear timescales has allowed DAMA/LIBRA to claim a statistically significant 9.3σ annual modulation with the expected period and phase of the standard DM halo [60]. However the parameter space favoured by these results ($7 \lesssim m_\chi \lesssim 15$ GeV/c^2) lies well within the exclusion limits of several experiments with much stronger optimisation of the four drivers, and correspondingly much lower σ_N across the DM mass range. This highlights that further work is needed to reconcile the results across all experimental classes.

4.4 Superheated Bubble Chamber

In bubble chambers, fluids (mostly refrigerants or halocarbons) are superheated in each operation cycle via initial compression followed by reduction of pressure [61], which lowers the fluid's boiling point such that it is below the temperature of the fluid. The fluid is metastable when superheated. Nucleation occurs when energy is deposited during a WIMP-nucleus interaction and a bubble is created. If the energy deposited does not exceed the energy threshold, the radius of the bubble is shorter than the critical radius, and the bubble collapses under the pressure of surface tension [61]. Otherwise, the bubble is captured by cameras in the detector, and acoustic signals are analysed for particle classification. In addition to SI interactions, SD interactions [61] can also be detected if fluids with non-zero nuclear spin, such as fluorocarbons (spin-$\frac{1}{2}$), are used.

Experiments of this type, such as PICO-60 and the emerging SBC collaboration, have a defining strength of near perfect rejection of electron-recoil backgrounds, making them a clear example of experiments that primarily optimise the background mitigation and suppression driver. Only nuclear recoils and α backgrounds deposit sufficient energy to form a bubble in the chamber, making the experiment effectively blind to ER [62]. On top of this, α backgrounds form distinct acoustic signatures that can be identified by piezoelectric transducers during bubble growth. Combined with low-radioactivity construction and strong neutron shielding, this yields background rejection factors $> 10^{10}$, unmatched amongst the other detection types discussed [63].

However this comes with significant trade-offs. Optimisation of the threshold driver is limited by a thermodynamic restriction on energy thresholds (constrained to $\mathcal{O}(3-10$ keV)) [64]. Larger volumes are also harder to keep thermodynamically metastable, as local perturbations in large tanks nucleate spontaneous bubbles. On top of this, the detectors have a dead-time after each event where the system returns to thermodynamic equilibrium. As a result, exposure is also difficult to scale with PICO-60 runs limited to 1400 kg-days [63].

The combination of target material that enables spin-dependant interactions, and near perfect background rejection, positions these types of detectors to be very competitive in the SD parameter space, for $m_\chi \approx 20-40$ GeV/c^2. However, the constraints on the other drivers mean that these experiments are unable to compete with the σ_N achieved by TPCs or cryogenic calorimeters in the broader SI parameter space.

4.5 Summary: Driver-Based Comparison of Detector Technologies

Table 1 summarises the results of the previous subsections. For each detector class analysed, we show which drivers are most strongly optimised, which are secondary, and which remain limiting. The final column maps these optimisation patterns onto the regions of the WIMP parameter space where each technology is currently most effective.

Table 1: Summary of the optimisation pattern of the technologies discussed in Subsections $4.1-4.4$, expressed in terms of the sensitivity drivers. Optimal mass range is accompanied by optimal target search function. Acronyms: LXe = liquid xenon; LAr = liquid argon; S2/S1 = charge-to-light ratio; PSD = pulse-shape discrimination; NR = nuclear recoil; ER = electronic recoil; SD = spin-dependent; SI = spin-independent.

Technology	**Exposure** (kg · yr)	**Target material**	**Background mitigation**	**Threshold** (keV)	**Optimal Mass Range** (GeV/c^2)
Noble-liquid TPCs	$\gtrsim 1000$	LXe / LAr	S2/S1, PSD, self-shielding	$\sim$ 1–5	SI, mid-high mass $m_\chi \gtrsim 10$
Cryogenic calorimeters	~ 1	multi-element targets	phonon and light/charge discrimination	$\sim$ 0.1–0.5	SI, low-mass $m_\chi \sim 0.3-10$
Scintillation crystals	~ 100	NaI targets	no NR/ER discrimination, intrinsic radioactivity	$\sim$ 10–20	annual-modulation tests, $m_\chi \sim 7-15$
Superheated bubble chambers	~ 1	SD targets	ER-blind, acoustic-α rejection	$\sim$ 3–10	SD-proton, $m_\chi \sim 5-50$

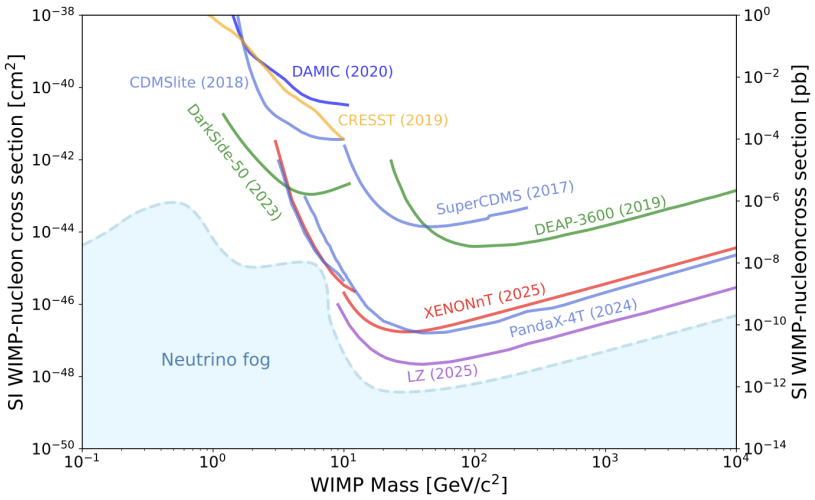


Figure 12: The current global exclusion limits on DM particle candidates, adopted from [65]

5 Discussion

The evolution of direct detection sensitivity, shown in Fig. 12, highlights a field whose progress is no longer uniform across the DM mass range. From 2018 to 2023, nearly all improvements occured in the low-mass region, while the latest results from LZ [16] set a world-leading sensitivity of $2.2 \times 10^{-48}\text{cm}^2$ at 40 GeV/c^2. The shape of the global exclusion limit shows different detector classes dominate in different regimes, highlighting fundamentally distinct drivers.

Noble Liquid TPCs achieve the strongest sensitivity in the $10-100$ GeV/c^2 region due scalable exposures and the A_T^2 coherence enhancement, whereas cryogenic detectors excel in the sub to few GeV/c^2 range due to energy thresholds three orders of magnitude times smaller than TPCs. As each technology faces a different limiting factor, background rejection for TPCs and phonon-noise limited scalability for cryogenic detectors, progress across the full mass range depends on optimising distinct physical limitations rather than the uniform improvement of one technology.

As TPC exposures continue to grow, their sensitivity will be constrained by increasingly demanding background rejection, particularly as they reach into the neutrino fog. Contrastingly, cryogenic detectors will encounter the fog at higher cross sections due to the spectral similarity between solar neutrinos and light DM recoils. Current analysis demonstrates cryogenic detectors ability to reach into the fog is dependent on exposures of $\mathcal{O}$(tonne-years) [66], yet existing detectors operate on $\mathcal{O}$(kilogram-year). This gap indicates that significant future engineering is required to increase the scalability of cryogenic detectors without compromising on energy thresholds or noise, whereas XLZD [67] is set to reach the fog with planned multi-tonne exposures.

Despite their differing strengths, both detector types are essential to the broader search strategy. Exclusion requires a null results with well-characterised backgrounds, but discovery requires a distinct signal that can be validated by multiple experiments. This is not only essential for signal confirmation but also for inferring DM particle properties. A single detector cannot determine the DM particle mass uniquely because the recoil energy depends on both mass and incident velocity. Different target materials provide different recoil response and therefore at least two complementary experiments are needed to constrain the particle mass and interaction strength [68]. This degeneracy structure is visible in Fig. 12, where overlapping sensitivities across targets enable confirmation of a future signal.

The merging of major collaborations such as Xenon, Lux-Zeplin and Darwin into XLZD reflects a natural consolidation as the field matures. Consolidation allows for improved discovery strength through the construction of larger scale experiments. However, consolidation also risks reducing competition and slowing adoption of innovative technologies, particularly those emerging from smaller experiments. Therefore, maintaining a balanced detector ensemble is crucial. Large collaborations allow for discovery-scale exposure whereas, smaller experiments provide the necessary degeneracy to confirm future signals.

Although current global limits rule out many simple DM candidates, a wide range of well motivated candidates remain [42]. A modern ensemble analysis, analogous to [68], performed on the current array of detectors would quantify the extent to which existing degeneracy is sufficient for confirmation of discovery and constraint of particle properties. Looking forward, the success of XLZD must be paired with competing high sensitivity experiments using alternative targets such as argon, as pursued by DarkSide-20k [69], to ensure any future candidate can be independently validated and its properties robustly constrained.

6 Conclusion

The purpose of this report was to compare the main experimental technologies in the search for DM by direct detection, and to analyse how their design choices determine achievable exclusion limits across the WIMP parameter space. The models we developed in Section 3, including our own toy model, allowed us to identify exposure, target material, recoil energy threshold, and background mitigation as the primary sensitivity drivers. While these models were highly simplified, even their limitations helped to highlight where real experiments face irreducible constraints.

Despite the simplifications, our comparison of experimental technologies showed the usefulness of the sensitivity-driver comparison framework for mapping how different detectors optimise different aspects of sensitivity. Noble-liquid TPCs maximise exposure and background mitigation, giving the strongest limit, $2.2 \times 10^{-48}\text{cm}^2$ for $m_\chi = 40$ GeV/c^2; cryogenic calorimeters minimise thresholds, dominating at low masses; scintillation crystals offer long-term stability for modulation searches; and bubble chambers provide optimal ER rejection for SD interactions. Through this analysis we were able to identify which specific detector improvements should be prioritised to improve global exclusion limits in each mass region.

The report shows that no technology or experiment alone can cover the entire mass range; the strength of the direct-detection field lies both in the optimisation of sensitivity drivers within each experimental class, and in the diversity of complementary technologies and approaches. Future progress will require a coordinated approach across detector technologies to probe the remaining DM parameter space and ensure that possible future signals can be independently detected and interpreted.

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